

JEE MAIN REVISION MASTER: COORDINATION COMPOUNDS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	IUPAC Nomenclature & Werner's Coordination Theory	25 PYQs	Coordination Basics
High	Crystal Field Theory (CFT) & Isomerism	10 PYQs	CFT & Isomerism

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
IUPAC Nomenclature & Werner's Coordination Theory	$\text{Primary valency (ionizable, satisfies charge)}, \quad \text{Secondary valency (non-ionizable, directional, satisfies CN)}$
Crystal Field Theory (CFT) & Isomerism	$\Delta_t = \frac{4}{9}\Delta_o, \quad \text{CFSE}_{\text{oct}} = [-0.4 n_{t_{2g}} + 0.6 n_{e_g}]\Delta_o + n_P, \quad \mu = \sqrt{n(n+2)} \text{ BM}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 IUPAC Nomenclature & Werner's Coordination Theory (Very High Yield — 25 PYQs)

Core Principle: Formulation of complexes, Werner's postulates, precipitable silver chloride fractions, and systematic IUPAC naming conventions.

■ **Caution / JEE Pitfall:** In $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$, adding excess AgNO_3 precipitates ONLY 2 moles of AgCl per mole of complex, because one Cl^- is coordinated.

■ **Speed Trick:** Chelating ligands (e.g. en, EDTA) form 5- or 6-membered rings with the central metal ion, imparting exceptional thermodynamic stability (chelate effect).

Representative Source PYQs:

- [2026 Shift] Q#3 (MCQ): "How many ions per molecule are produced from the complex $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$ in solution?..." [Ans: N/A]
- [2026 Shift] Q#4 (MCQ): "Which of the following is the most stable complex ion?..." [Ans: B]
- [2025 Shift] Q#1 (MCQ): "A ligand which has two different donor atoms and either of the two ligates with the central metal atom/ion in the complex is called _____." [Ans: D]

3.2 Crystal Field Theory (CFT) & Isomerism (High Yield — 10 PYQs)

Core Principle: d-orbital splitting under octahedral and tetrahedral fields, high-spin vs low-spin complexes, spectrochemical series, and optical/geometrical isomerism.

■ **Caution / JEE Pitfall:** Tetrahedral complexes NEVER form low-spin complexes because $\Delta_t = (4/9)\Delta_o < P$ (splitting is always less than pairing energy).

■ **Speed Trick:** cis-[Co(en)₂Cl₂]⁺ is optically ACTIVE (non-superimposable mirror image), whereas the trans isomer has a plane of symmetry and is optically INACTIVE.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Match List-I with List-II List-I (Complex) List-II (Geometry) (a) [Co(NH₃)₆]³⁺ (i) Trigonal bipyramidal (b) [NiCl₄]²⁻ (ii) Octahedral (c) [Ni(CN)₄]²⁻ (iii) Tetrahedral (d) [Fe(CO)₅] (iv) Square planar Choose the correct answer from the options given below." **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "Given below are two statements: Statement-I: The M-C σ bond is formed by the donation of lone pair of electrons on th..." **[Ans: N/A]**
- **[2026 Shift] Q#4 (MCQ):** "Which of the following is the most stable complex ion?..." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match List-I with List-II List-I (Complex) List-II (Geometry) (a) [Co(NH₃)₆]³⁺ (i) Trigonal bipyramidal (b) [NiCl₄]²⁻ (ii) Octahedral (c) [Ni(CN)₄]²⁻ (iii) Tetrahedral (d) [Fe(CO)₅] (iv) Square planar Choose the correct answer from the options given below.

- (A) a-ii, b-iii, c-iv, d-i
- (B) a-ii, b-i, c-iii, d-iv
- (C) a-iii, b-ii, c-iv, d-i
- (D) a-i, b-iii, c-iv, d-ii

Correct Answer: None

Solution Analysis: For each complex, we identify the coordination number and the usual geometry. [Co(NH₃)₆]³⁺ has 6 ligands around cobalt. So its coordination number is 6, and complexes with coordination number 6 generally have octahedral geometry. (a) → (ii) [NiCl₄]²⁻ has 4 chloride ligands around nickel. Here nickel is in +2 oxidation state, so it is a d⁸ ion. Since Cl⁻ is a weak field ligand, the complex remains tetrahedral. (b) → (iii) [Ni(CN)₄]²⁻ also has nickel in +2 oxidation state, so it is again d⁸. B

Benchmark Problem 2: KCET 2026 (Q#3)

How many ions per molecule are produced from the complex [Co(NH₃)₆]Cl₂ in solution?

- (A) 6
- (B) 4
- (C) 3
- (D) 2

Correct Answer: None

Solution Analysis: In [Co(NH₃)₆]Cl₂, the two chloride ions are outside the coordination sphere. So in water, it dissociates as: [Co(NH₃)₆]Cl₂ → [Co(NH₃)₆]²⁺ + 2Cl⁻. The complex ion [Co(NH₃)₆]²⁺ remains as one ion, and the two chloride ions are separate. So total ions produced: 1+2=3 Hence, the correct option is: Option C3